

$$\begin{aligned}
& \triangleright f(b, c, r) := \left(\frac{1}{2} - \cos\left(\frac{5}{6} \pi - c - r\right) \cdot \sin(c) - \cos\left(\frac{5}{6} \pi - b + r\right) \cdot \sin(b) \right)^2 + \left(\sin\left(\frac{5}{6} \pi - c - r\right) \cdot \sin(c) - \sin\left(\frac{5}{6} \pi - b + r\right) \cdot \sin(b) \right)^2 \\
& f := (b, c, r) \mapsto \left(\frac{1}{2} - \cos\left(\frac{5 \cdot \pi}{6} - c - r\right) \cdot \sin(c) - \cos\left(\frac{5 \cdot \pi}{6} - b + r\right) \cdot \sin(b) \right)^2 + \left(\sin\left(\frac{5 \cdot \pi}{6} - c - r\right) \cdot \sin(c) - \sin\left(\frac{5 \cdot \pi}{6} - b + r\right) \cdot \sin(b) \right)^2 \quad (1)
\end{aligned}$$

$$\begin{aligned}
& \triangleright g(b, c, r) := \text{simplify}(\text{expand}(f(b, c, 0)^2 - f(b, c, r) \cdot f(b, c, -r))) \\
& \quad g := (b, c, r) \mapsto \text{simplify}(\text{expand}(f(b, c, 0)^2 - f(b, c, r) \cdot f(b, c, -r))) \quad (2)
\end{aligned}$$

$$\begin{aligned}
& \triangleright \text{collect}(\text{expand}(g(b, c, r)), \cos(r)) \\
& (\sin(c)^2 \sin(b) \sqrt{3} \cos(b) + \sin(c) \sqrt{3} \cos(c) \sin(b)^2 - \sin(c) \cos(c) \sin(b) \cos(b) \quad (3)
\end{aligned}$$

$$\begin{aligned}
& + \cos(b)^2 \cos(c)^2 - 1) \cos(r)^2 + \left(-4 \sin(c) \cos(c) \sin(b) \cos(b)^3 \right. \\
& + 2 \sin(c) \sqrt{3} \cos(c) \cos(b)^2 + 2 \sqrt{3} \sin(b) \cos(b) \cos(c)^2 + 8 \sin(c) \cos(c) \sin(b) \cos(b) \\
& - 4 \sin(c) \cos(c)^3 \sin(b) \cos(b) - 3 \cos(c)^4 + \frac{7 \cos(c)^2}{4} - 3 \cos(b)^4 + \frac{7 \cos(b)^2}{4} + \frac{5}{2} \\
& + \sin(c) \sqrt{3} \cos(c)^3 - \frac{13 \sin(c) \sqrt{3} \cos(c)}{4} + \sqrt{3} \sin(b) \cos(b)^3 \\
& \left. - \frac{13 \sin(b) \sqrt{3} \cos(b)}{4} - 8 \cos(b)^2 \cos(c)^2 + 4 \cos(b)^2 \cos(c)^4 + 4 \cos(c)^2 \cos(b)^4 \right) \\
& \cos(r) + 4 \sin(c) \cos(c) \sin(b) \cos(b)^3 - 7 \sin(c) \cos(c) \sin(b) \cos(b) \\
& - \sin(c) \sqrt{3} \cos(c) \cos(b)^2 - \sqrt{3} \sin(b) \cos(b) \cos(c)^2 + 4 \sin(c) \cos(c)^3 \sin(b) \cos(b) \\
& - \frac{7 \cos(b)^2}{4} - \frac{7 \cos(c)^2}{4} + 3 \cos(c)^4 + 3 \cos(b)^4 - \sin(c) \sqrt{3} \cos(c)^3 \\
& + \frac{9 \sin(c) \sqrt{3} \cos(c)}{4} - \sqrt{3} \sin(b) \cos(b)^3 + \frac{9 \sin(b) \sqrt{3} \cos(b)}{4} - 4 \cos(b)^2 \cos(c)^4 \\
& - 4 \cos(c)^2 \cos(b)^4 + 7 \cos(b)^2 \cos(c)^2 - \frac{3}{2}
\end{aligned}$$

$$\begin{aligned}
& \triangleright \text{simplify} \left(\frac{1}{r^2} \left((\sin(c)^2 \sin(b) \sqrt{3} \cos(b) + \sin(c) \sqrt{3} \cos(c) \sin(b)^2 \right. \right. \\
& \quad \left. \left. - \sin(c) \cos(c) \sin(b) \cos(b) + \cos(b)^2 \cos(c)^2 - 1) \cdot (1 - r^2) + \left(\right. \right. \right. \\
& \quad \left. \left. - 4 \sin(c) \cos(c) \sin(b) \cos(b)^3 + 2 \sin(c) \sqrt{3} \cos(c) \cos(b)^2 \right. \right.
\end{aligned}$$

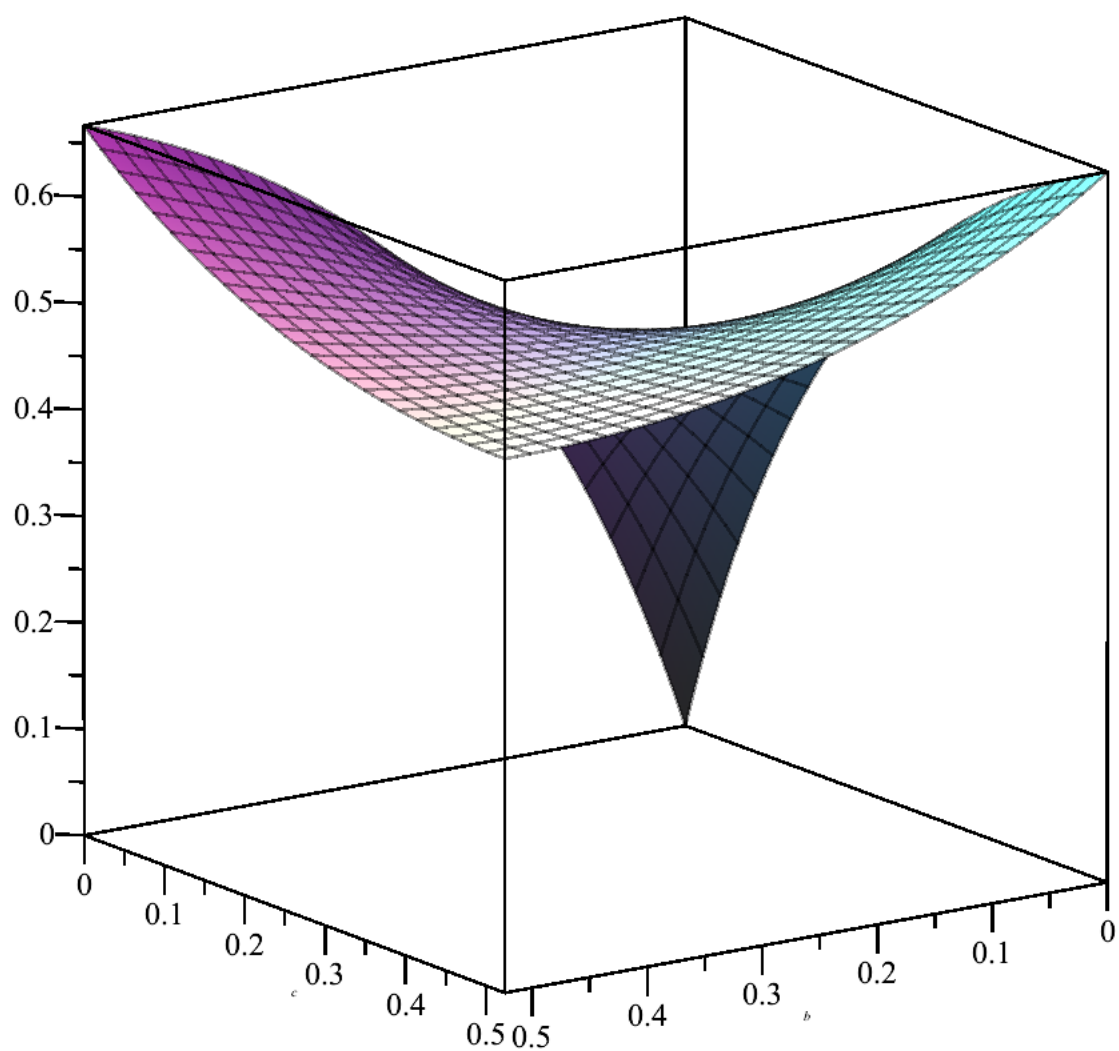
$$\begin{aligned}
& + 2\sqrt{3} \sin(b) \cos(b) \cos(c)^2 + 8 \sin(c) \cos(c) \sin(b) \cos(b) \\
& - 4 \sin(c) \cos(c)^3 \sin(b) \cos(b) - 3 \cos(c)^4 + \frac{7 \cos(c)^2}{4} - 3 \cos(b)^4 + \frac{7 \cos(b)^2}{4} \\
& + \frac{5}{2} + \sin(c) \sqrt{3} \cos(c)^3 - \frac{13 \sin(c) \sqrt{3} \cos(c)}{4} + \sqrt{3} \sin(b) \cos(b)^3 \\
& - \frac{13 \sin(b) \sqrt{3} \cos(b)}{4} - 8 \cos(b)^2 \cos(c)^2 + 4 \cos(b)^2 \cos(c)^4 + 4 \cos(c)^2 \cos(b)^4 \Big) \\
& \cdot \left(1 - \frac{r^2}{2} \right) + 4 \sin(c) \cos(c) \sin(b) \cos(b)^3 - 7 \sin(c) \cos(c) \sin(b) \cos(b) \\
& - \sin(c) \sqrt{3} \cos(c) \cos(b)^2 - \sqrt{3} \sin(b) \cos(b) \cos(c)^2 + 4 \sin(c) \cos(c)^3 \sin(b) \cos(b) \\
& - \frac{7 \cos(b)^2}{4} - \frac{7 \cos(c)^2}{4} + 3 \cos(c)^4 + 3 \cos(b)^4 - \sin(c) \sqrt{3} \cos(c)^3 \\
& + \frac{9 \sin(c) \sqrt{3} \cos(c)}{4} - \sqrt{3} \sin(b) \cos(b)^3 + \frac{9 \sin(b) \sqrt{3} \cos(b)}{4} - 4 \cos(b)^2 \cos(c)^4 \\
& - 4 \cos(c)^2 \cos(b)^4 + 7 \cos(b)^2 \cos(c)^2 - \frac{3}{2} \Big) \Big) \\
& - \frac{1}{4} + \frac{(-4 \sin(b) \cos(b)^3 - 4 \sin(c) \cos(c)^3 + 5 \sin(b) \cos(b) + 5 \cos(c) \sin(c)) \sqrt{3}}{8} \\
& + \frac{(-4 \cos(c)^2 + 3) \cos(b)^4}{2} + 2 \sin(c) \cos(c) \sin(b) \cos(b)^3 \\
& + \frac{(-16 \cos(c)^4 + 24 \cos(c)^2 - 7) \cos(b)^2}{8} + 2 \sin(b) \left(\cos(c)^2 \right. \\
& \left. - \frac{3}{2} \right) \cos(c) \sin(c) \cos(b) + \frac{3 \cos(c)^4}{2} - \frac{7 \cos(c)^2}{8}
\end{aligned} \tag{4}$$

$$\begin{aligned}
> \quad h(b, c) &:= -\frac{1}{4} \\
& + \frac{(-4 \sin(b) \cos(b)^3 - 4 \sin(c) \cos(c)^3 + 5 \sin(b) \cos(b) + 5 \cos(c) \sin(c)) \sqrt{3}}{8} \\
& + \frac{(-4 \cos(c)^2 + 3) \cos(b)^4}{2} + 2 \sin(c) \cos(c) \sin(b) \cos(b)^3 \\
& + \frac{(-16 \cos(c)^4 + 24 \cos(c)^2 - 7) \cos(b)^2}{8} + 2 \sin(b) \left(\cos(c)^2 \right. \\
& \left. - \frac{3}{2} \right) \cos(c) \sin(c) \cos(b) + \frac{3 \cos(c)^4}{2} - \frac{7 \cos(c)^2}{8}
\end{aligned}$$

$$\begin{aligned}
h &:= (b, c) \mapsto -\frac{1}{4} \\
& + \frac{(-4 \cdot \sin(b) \cdot \cos(b)^3 - 4 \cdot \sin(c) \cdot \cos(c)^3 + 5 \cdot \sin(b) \cdot \cos(b) + 5 \cdot \cos(c) \cdot \sin(c)) \cdot \sqrt{3}}{8}
\end{aligned} \tag{5}$$

$$\begin{aligned}
& + \frac{(-4 \cdot \cos(c)^2 + 3) \cdot \cos(b)^4}{2} + 2 \cdot \cos(c) \cdot \sin(c) \cdot \sin(b) \cdot \cos(b)^3 \\
& + \frac{(-16 \cdot \cos(c)^4 + 24 \cdot \cos(c)^2 - 7) \cdot \cos(b)^2}{8} + 2 \cdot \sin(b) \cdot \left(\cos(c)^2 - \frac{3}{2} \right) \cdot \cos(c) \cdot \sin(c) \\
& \cdot \cos(b) + \frac{3 \cdot \cos(c)^4}{2} - \frac{7 \cdot \cos(c)^2}{8}
\end{aligned}$$

> $\text{plot3d}\left(\frac{h(b, c)}{f(c, b, 0)^2}, b=0..\frac{3.14}{6}, c=0..\frac{3.14}{6}\right)$



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